Smart Queue Management System: Software Requirements Specification

Milestone 2 (Session 2): Requirements Engineering and System Analysis

Course: Software Engineering and User-Centred Design | Team: T6 | Tag: m2 | Based on M1 pain points PP1 to PP5

1. Introduction

1.1 Purpose

This SRS states precisely what the Smart Queue Management System (SQMS) must do so that another team could build it without asking us to explain. It follows an IEEE 29148-inspired outline [R1] and traces every requirement to a pain point found in Milestone 1 interviews in Rwamagana (P1 to P5).

1.2 Scope

In scope (this course)	Out of scope (this course)
Customer ticket request at a branch device (tablet web page) using a phone number and service type Automatic sequential queue numbers SMS: confirmation with estimated wait, turn-near alert, called-to-counter alert (SMS gateway simulated, for example Africa's Talking sandbox) Priority queue for elderly, pregnant and disabled customers per institution policy Teller screen: call next, recall, mark served or no-show Manager dashboard and simple report export Kinyarwanda and English text Staff login with roles; one branch	Integration with core banking systems Appointment booking and online payments Native mobile app Manufacture of physical kiosk or ticket-printer hardware Biometric or national ID verification Multi-branch head-office analytics Live commercial SMS contracts and production deployment

1.3 Definitions

Term	Meaning
SQMS	Smart Queue Management System
Ticket	A queue number issued to one customer for one service
Priority customer	Elderly person, pregnant woman or person with a disability, as defined by the institution's priority policy
Estimated wait (ETW)	Number of customers ahead multiplied by the recent average service time, shown in minutes
No-show	Customer who does not reach the counter within the grace period after being called
SRS / FR / NFR	Software Requirements Specification / Functional / Non-functional Requirement

2. Overall Description

2.1 Users and personas

Actor	Persona (M1)	Main needs from the system
Customer	Eric (P1), Grace (P3)	Know position and wait; leave the hall and return; fair order (PP1, PP3)
Priority customer	Mama Rose (P2)	Be called earlier without asking; avoid long standing (PP2)
Teller	Claudine (P4)	See who is next and call them; fewer interruptions (PP3, PP4)
Branch manager	Jean Paul (P5)	Live queue view, wait times, peak hours (PP5)
Administrator	IT support	Configure services, counters and thresholds

Actor	Persona (M1)	Main needs from the system
SMS gateway (external)	SMS providers	Deliver messages reliably

2.2 Operating environment

- Customer device: tablet (about 10 inch) in the banking hall running a responsive web page; tellers and managers use desktop or laptop browsers (latest two versions of Chrome or Edge).
- Back end: REST API with a relational database; SMS sent through a gateway (sandbox during the course).
- Customers receive SMS on any mobile phone; no smartphone or data bundle is needed.

2.3 Constraints

- Two-to-three-person team; Sessions 4 and 5 are the only build and test time, so Must stories must fit within them.
- No coding before design approval at the end of Session 3; design-freeze applies to Must scope.
- Phone numbers are personal data and must be handled under Rwanda's data-protection law [R2].

2.4 Assumptions

- Branches have a stable power supply and a network connection during opening hours.
- Each institution can supply its own priority policy and service types.
- Most customers own a mobile phone; those without one are handled by the floor guide (FR-18).
- Device and connectivity details of personas were not recorded in M1 and will be confirmed during user validation (Section 9).

3. Functional Requirements

ID	Requirement	Priority	Source pain point
FR-01	The system shall let a customer request a ticket by selecting a service type and entering a mobile phone number on the branch device.	Must	PP1, PP3
FR-02	The system shall display a consent notice explaining that the phone number is used only for queue SMS, and require acceptance before issuing a ticket.	Must	Privacy [R2]
FR-03	The system shall generate a unique sequential queue number per branch, service and day automatically.	Must	PP3
FR-04	The system shall send an SMS confirming the queue number, service and estimated wait within seconds of ticket issue.	Must	PP1
FR-05	The system shall calculate the estimated wait from the number of customers ahead and the recent average service time, and update it as the queue moves.	Must	PP1
FR-06	The system shall send an SMS when a configurable number of customers (default 3) remain ahead of the customer.	Must	PP1
FR-07	The system shall send an SMS naming the counter when a customer is called.	Should	PP1
FR-08	The system shall let a customer who is eligible choose the priority category (elderly, pregnant, person with disability) and shall place priority tickets ahead of standard tickets per the institution's policy.	Must	PP2
FR-09	The system shall let a teller view the waiting list for their service and call the next ticket (priority first, then first-come-first-served).	Must	PP3, PP4
FR-10	The system shall let a teller mark a ticket as served, recall a called ticket, or mark it as no-show after the grace period (default 5 minutes).	Must	PP3, PP4
FR-11	The system shall keep a customer's place while they are away from the hall until they are called, and let a no-show customer rejoin at the next available position once.	Must	PP1
FR-12	The system shall record timestamps for ticket created, called, service started and completed.	Must	PP5

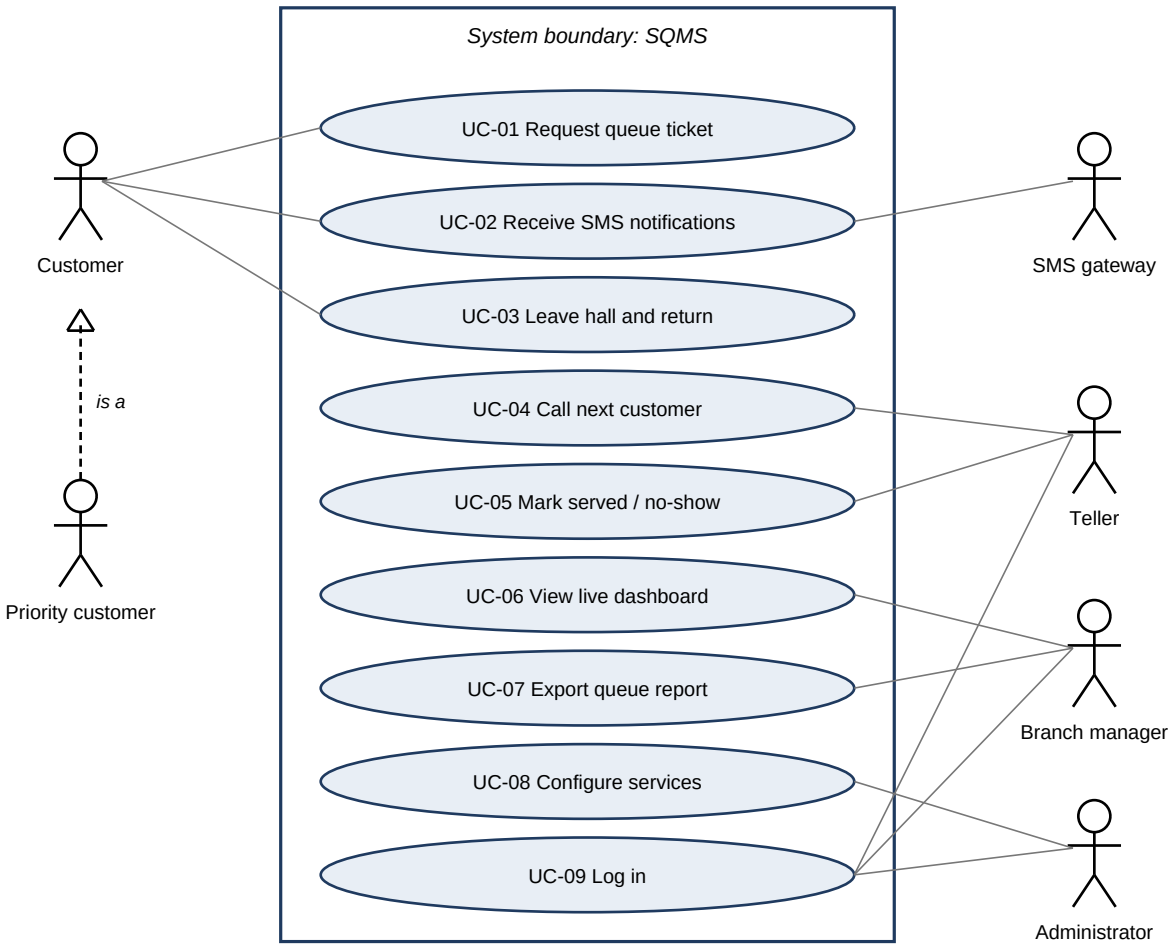
ID	Requirement	Priority	Source pain point
FR-13	The system shall give a manager a live dashboard: customers waiting, average wait, customers served and busiest hours for the current day.	Must	PP5
FR-14	The system shall let a manager export queue statistics by day and hour as CSV.	Should	PP5
FR-15	The system shall present kiosk screens and SMS texts in Kinyarwanda and English, selectable by the customer.	Should	PP1, PP2
FR-16	The system shall require staff login with roles (teller, manager, administrator) and restrict functions by role.	Must	PP4, PP5
FR-17	The system shall let an administrator configure services, counters, priority rules and notification thresholds.	Should	PP2, PP5
FR-18	The system shall let the floor guide or teller issue a ticket for a customer without a phone number; the ticket number is shown on the hall display.	Should	PP2
FR-19	The system shall show currently served numbers and counters on a hall display screen.	Could	PP3
FR-20	The system shall let a customer reply to an SMS with a keyword to receive the current status of their ticket.	Could	PP1

4. Non-Functional Requirements

ID	Category	Measurable criterion
NFR-01	Performance	A ticket is issued and shown on the kiosk within 3 seconds for 95% of requests with 50 customers using the system at once.
NFR-02	Performance	An SMS is handed to the gateway within 10 seconds of the triggering event for 95% of messages.
NFR-03	Performance	The teller screen and manager dashboard refresh within 2 seconds of a queue change.
NFR-04	Availability	The system is available at least 99% of branch opening hours (08:00 to 17:00) per month.
NFR-05	Privacy	Phone numbers are collected only with consent (FR-02), shown masked to staff (last 3 digits), and deleted or anonymised after 30 days. Design follows consent and purpose-limitation principles of Law No. 058/2021 [R2]; legal review pending.
NFR-06	Security	All traffic uses TLS 1.2 or higher; staff passwords are stored hashed; sessions expire after 15 minutes of inactivity; access is role-based.
NFR-07	Usability	At least 80% of first-time customers complete ticket registration in under 60 seconds without help (tested in M5 with 3+ users).
NFR-08	Accessibility	Text contrast at least 4.5:1, touch targets at least 44 px, a label on every input; web screens target WCAG 2.2 level AA [R3].
NFR-09	Localisation	All customer-facing text exists in Kinyarwanda and English; each SMS fits one 160-character segment.
NFR-10	Accuracy	Estimated wait is within 10 minutes of actual wait for at least 80% of tickets in simulated test runs.
NFR-11	Scalability	One branch supports at least 500 tickets per day and 5 counters without degradation of NFR-01.
NFR-12	Data integrity	No two tickets share the same number for the same branch, service and day, including under concurrent requests.
NFR-13	Compatibility	Staff screens work at 1366 x 768 resolution on the latest two versions of Chrome and Edge; kiosk page works on a 10 inch Android tablet.
NFR-14	Maintainability	Every API endpoint is documented; at least 5 automated tests run on every push (M5).

5. Use Case Diagram and Detailed Use Cases

SQMS use case diagram



Editable source: T6-M2-UseCase.drawio (re-draw this diagram in Draw.io and attach the .drawio file and PNG export with the submission).

UC-01 Request queue ticket (main flow and alternative flows)

Actor	Customer (and Priority customer); SMS gateway as secondary actor
Precondition	Kiosk page is running; branch is open; at least one service is configured
Main flow	1. Customer selects a language and a service. 2. Customer enters a mobile phone number. 3. System shows the consent notice; customer accepts. 4. System issues the next sequential ticket and calculates the estimated wait (FR-03, FR-05). 5. System shows the ticket number, position and wait on screen. 6. System sends the confirmation SMS (FR-04). 7. Customer may leave the hall and return when alerted.
Alternative flows	A1 Priority customer: at step 1 the customer chooses a priority category; the ticket is placed ahead of standard tickets (FR-08). A2 Invalid phone number: system shows an error in the chosen language and asks again. A3 Consent refused: no ticket is issued; the system suggests asking the floor guide (FR-18). A4 SMS fails: the ticket remains valid and its number stays on screen; system retries and logs the failure.
Postcondition	A ticket exists with status Waiting and a created timestamp (FR-12)

UC-04 Call next customer

Actor	Teller; SMS gateway as secondary actor
Precondition	Teller is logged in and has opened a counter; at least one ticket is Waiting

Main flow	1. Teller presses Call next. 2. System selects the first priority ticket, or if none the oldest standard ticket (FR-09). 3. System sets the ticket to Called, records the called time, updates the hall display and sends the called-to-counter SMS (FR-07). 4. Customer reaches the counter; teller starts service. 5. Teller presses Served; system records completion time and updates the average service time (FR-10, FR-12).
Alternative flows	A1 No-show: after the grace period the teller marks No-show; the ticket is skipped and the customer may rejoin once (FR-11). A2 Recall: teller recalls a Called ticket and the SMS and display are repeated. A3 Empty queue: system shows 'No customers waiting'.
Postcondition	Ticket status is Served or No-show; queue statistics are updated for the dashboard (FR-13)

6. User Stories with Acceptance Criteria

Stories live in Jira (or Trello): add the board link and a screenshot with the submission. Story points use the Fibonacci-style scale; priorities use MoSCoW.

US-01 As a customer, I want to request a queue ticket at the branch by choosing a service and entering my phone number so that I do not need an unreliable paper ticket and my place is recorded.

Priority: Must | **Story points:** 5 | **Traces to:** PP1, PP3, FR-01, FR-02, FR-03

Acceptance criteria
Given I am at the kiosk and have accepted the consent notice
When I choose 'Withdrawal' and enter a valid phone number
Then I see my queue number, position and estimated wait within 3 seconds, and the number is higher than the last one issued for that service.

US-02 As a customer, I want to receive an SMS with my queue number and estimated wait so that I can plan my time without standing in the hall.

Priority: Must | **Story points:** 3 | **Traces to:** PP1, FR-04, FR-05

Acceptance criteria
Given I have a ticket
When it is issued
Then I receive an SMS within 10 seconds containing my number and estimated wait in minutes, in the language I chose.

US-03 As a customer, I want to receive an SMS when my turn is near so that I can leave the hall and return just before I am called.

Priority: Must | **Story points:** 3 | **Traces to:** PP1, FR-06, FR-11

Acceptance criteria
Given I have a ticket and only 3 customers are ahead of me
When the third customer before me is called
Then I receive an SMS telling me to come to the branch, and my place remains valid while I am away.

US-04 As a priority customer (elderly, pregnant, disabled), I want to get a priority ticket without having to ask staff so that I am served earlier and do not stand for a long time.

Priority: Must | **Story points:** 3 | **Traces to:** PP2, FR-08

Acceptance criteria
Given I select a priority category at the kiosk
When I submit my phone number
Then my ticket is marked Priority and is called before standard tickets that arrived after it, following the institution's policy.

US-05 As a teller, I want to call the next customer with one action so that I serve in a clear, fair order and stop answering 'who is next?'.

Priority: Must | **Story points:** 5 | **Traces to:** PP3, PP4, FR-09, FR-10

Acceptance criteria
Given I am logged in and customers are waiting
When I press Call next
Then the first priority ticket (or oldest standard ticket) becomes Called, appears on my screen and the hall display, and the customer is notified by SMS.

US-06 As a teller, I want to mark a customer as served or no-show and recall a called ticket so that the queue keeps moving and disputes are avoided.

Priority: Must | **Story points:** 3 | **Traces to:** PP3, PP4, FR-10, FR-11

Acceptance criteria
Given a ticket is Called and the customer has not arrived after 5 minutes
When I press No-show
Then the ticket is skipped, the next ticket can be called, and the customer can rejoin once.

US-07 As a branch manager, I want to see the live queue and today's waiting times so that I can react when the hall becomes crowded.

Priority: Must | **Story points:** 5 | **Traces to:** PP5, FR-12, FR-13

Acceptance criteria
Given I am logged in as manager
When I open the dashboard
Then I see customers waiting, average wait, customers served and busiest hours, refreshed within 2 seconds of a change.

US-08 As a staff member, I want to log in with a role so that only authorised people can call tickets or view reports.

Priority: Must | **Story points:** 3 | **Traces to:** PP4, PP5, FR-16

Acceptance criteria
Given I am a teller
When I try to open the manager dashboard
Then access is denied; and given I enter a wrong password three times, then the account is locked for 5 minutes.

US-09 As a branch manager, I want to export queue statistics by day and hour so that I can plan staffing for peak periods.

Priority: Should | **Story points:** 3 | **Traces to:** PP5, FR-14

Acceptance criteria
Given one week of ticket data exists
When I choose Export
Then I receive a CSV with tickets, average wait and customers served per hour.

US-10 As a customer, I want to use the kiosk and SMS in Kinyarwanda or English so that I understand every message.

Priority: Should | **Story points:** 2 | **Traces to:** PP1, PP2, FR-15

Acceptance criteria
Given I choose Kinyarwanda on the first screen
When I complete registration
Then every screen and SMS is in Kinyarwanda.

US-11 As a customer without a phone, I want to get a ticket from the floor guide so that I am not excluded from the queue system.

Priority: Should | **Story points:** 3 | **Traces to:** PP2, FR-18

Acceptance criteria
Given I have no mobile phone
When the floor guide issues a ticket without a number
Then the ticket number appears on the hall display when I am near the front.

US-12 As a administrator, I want to configure services, counters and the 'turn near' threshold so that the system fits each branch's policy.
Priority: Should | **Story points:** 3 | **Traces to:** PP2, PP5, FR-17
Acceptance criteria Given I am logged in as administrator When I change the threshold from 3 to 5 Then new 'turn near' SMS messages are sent when 5 customers remain ahead.

US-13 As a customer, I want to reply STATUS to an SMS to learn my position so that I can check without returning to the branch.
Priority: Could | **Story points:** 3 | **Traces to:** PP1, FR-20
Acceptance criteria Given I hold a Waiting ticket When I reply STATUS Then I receive my current position and estimated wait.

US-14 As a customer in the hall, I want to see current numbers on a display screen so that I can follow progress with my own eyes.
Priority: Could | **Story points:** 2 | **Traces to:** PP3, FR-19
Acceptance criteria Given counters are serving tickets When a ticket is called Then the hall display shows the number and counter within 2 seconds.

7. MoSCoW Prioritised Backlog

Priority	Stories	Total points
Must (core of Sessions 4 and 5)	US-01, US-02, US-03, US-04, US-05, US-06, US-07, US-08	30
Should	US-09, US-10, US-11, US-12	11
Could	US-13, US-14	5
Won't (this course)	Core-banking integration, appointment booking, mobile app, payments, multi-branch analytics	-

The Must set (30 points) is a realistic target for two build sprints; if velocity is lower, US-07 can be reduced to a basic queue-length view without removing it.

8. Traceability Matrix: Pain Point to Requirement to User Story

Pain point (M1)	Functional requirements	Non-functional requirements	User stories
PP1 Long, unpredictable waiting; cannot leave queue	FR-01, FR-04, FR-05, FR-06, FR-07, FR-11, FR-15, FR-20	NFR-01, NFR-02, NFR-09, NFR-10	US-01, US-02, US-03, US-10, US-13
PP2 No seating or priority for vulnerable customers	FR-08, FR-15, FR-17, FR-18	NFR-07, NFR-08	US-04, US-10, US-11, US-12
PP3 Unclear order and disputes	FR-01, FR-03, FR-09, FR-10, FR-19	NFR-03, NFR-12	US-01, US-05, US-06, US-14
PP4 Staff workload and interruptions	FR-09, FR-10, FR-16	NFR-03, NFR-06	US-05, US-06, US-08
PP5 No queue data for management	FR-12, FR-13, FR-14, FR-16, FR-17	NFR-04, NFR-05, NFR-11	US-07, US-08, US-09, US-12
Privacy and legal (all users)	FR-02	NFR-05, NFR-06	US-01

9. User Validation Notes (2 users)

To be completed by the team. Show the Must-have list below to two real users (suggested: one customer or priority customer from the M1 interviewees, and one teller or manager), ask what they would change, and record the answers. These cells are intentionally left blank: they must contain real feedback, not assumptions.

Must story	User A (role: ____) feedback	User B (role: ____) feedback	Change made to requirement
US-01 request a queue ticket at the branch by choosing...			
US-02 receive an SMS with my queue number and estimate...			
US-03 receive an SMS when my turn is near			
US-04 get a priority ticket without having to ask staf...			
US-05 call the next customer with one action			
US-06 mark a customer as served or no-show and recall ...			
US-07 see the live queue and today's waiting times			
US-08 log in with a role			

Also record: date, participants (anonymised as V1 and V2), language used, and any new requirement that results (add it to the FR table with a new ID).

10. Milestone 2 Checklist

Deliverable	Where	Status
SRS document (PDF)	This file	Done
Scope statement	Section 1.2	Done
Numbered functional and non-functional requirements	Sections 3 and 4	Done
8+ user stories with Given/When/Then in Jira or Trello (link + screenshot)	Section 6 (14 stories)	Create board, add link and screenshot
MoSCoW-prioritised backlog	Section 7	Done
Use case diagram (.drawio + PNG)	Section 5	Re-draw in Draw.io and attach files
Traceability matrix	Section 8	Done
User validation notes from 2 users	Section 9	To be completed with real users
sources.md with 3+ cited sources; ai-use-log.md updated	Sources below; docs/	Update in repository
GitHub release tag m2; link on class platform	Repository	Pending

Approval question: could another development team build from these requirements without asking us to explain them? Each requirement is numbered, testable and traced to a pain point; open items are the real user validation and the editable diagram and board links.

Sources

[R1] International Organization for Standardization, International Electrotechnical Commission, & Institute of Electrical and Electronics Engineers. (2018). *ISO/IEC/IEEE 29148:2018 Systems and software engineering: Life cycle processes: Requirements engineering*. ISO. https://www.iso.org/standard/72089.html

[R2] Republic of Rwanda. (2021). *Law No. 058/2021 of 13/10/2021 relating to the protection of personal data and privacy*. Official Gazette of the Republic of Rwanda, Special issue of 15/10/2021. (Supervisory authority: National Cyber Security Authority. Add the official gazette PDF link in sources.md.)

[R3] World Wide Web Consortium. (2023). *Web Content Accessibility Guidelines (WCAG) 2.2*. W3C. https://www.w3.org/TR/WCAG22/

M1 sources [S1] to [S9] are listed in T6-M1-ProblemDiscovery.pdf and docs/research/sources.md.